

Human health risk assessment (HHRA) of fluoride and nitrate using pollution index of groundwater (PIG) in and around hard rock terrain of Cuddapah, A.P. South India

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ABSTRACT

Groundwater is the most common source of drinking and irrigation in the world. As a result of human activities and natural processes, nitrate levels in most groundwater supplies in arid and semi-arid areas have risen in recent decades. As a result of human activities and natural processes, the levels of fluoride and nitrate in most groundwater supplies in arid and semi-arid areas have increased in recent decades. Thirty groundwater samples were taken from bore wells to learn about the contaminated situations of groundwater fluoride and nitrate in sections of Cuddapah, South India, and to determine the likely harm to the health of adults, children, and infants. In the study villages' wells, fluoride concentrations range from 0.1–3.2 mg/L, with 40% of groundwater exceeding fluoride guidelines (1.5 mg/L) and Nitrate concentrations range from 23.2–110.8 mg/L, with 86.6% of groundwater exceeding nitrate guidelines (45 mg/L). 96% of groundwater exceeds the allowable THI values (>1) for adults (0.74–4.02), children (0.75–4.08), and infants (0.57–3.10). According to the PIG categorization, 23% of total groundwater samples fall into the low pollution zone (PIG: 1.38) whereas 47% of samples fall into the moderate pollution zone (1.83) and 30% of groundwater samples fall into the high pollution zone (2.20). Hence, appropriate preventative steps must be adopted to reduce the health risk in this area.

1. Introduction

Groundwater constitutes an important onset of fresh water for drinking and agriculture in the world's arid and semi-arid regions. In the world, more than 1.5 billion people entrust on groundwater for their drinking water. Groundwater is the primary source for meeting the needs of rural residents for daily activities, particularly in a developing country like India [1,2]. In dry and semi-dry parts of various countries, groundwater is the primary replenishable supply of fresh water for residential, agriculture, and industrial needs. Drinking water accounts for 65% of groundwater use globally, farming and animal feeding accounts for 20%, while industry and mining accounts for roughly 15% of groundwater use [3]. Quality of groundwater in the rain shadow zone of semi-arid regions is characteristically a function of regional geo-hydrology, lithology, evapotranspiration, rainfall quality, and quantity, as well as host rock reactions with precipitation, inorganic compound solubilities, aquifer mineralogy, and weathering characteristics.

Anthropogenic factors such as industrial processes and wastes, the release of fertilizers, herbicides, and animal manure lower beneficial uses of groundwater resources [4–6]. Human health and water quality are inextricably linked. Water pollution by fluoride and nitrate is a serious public health concern around the world for the past two decades. These research works were carried out in arid and semi-arid locations everywhere in the globe due to their importance [7–13].

Fluoride levels are higher in greater than 23 nations globally, especially in India, China, Central Africa, and South America. Several groundwater quality studies with fluoride were conducted in India's arid and semi-arid regions. Even though fluoride is necessary for the natural mineralization of bones and the creation of dental enamel in modest amounts, higher concentration levels (>1.5 mg/L) produce dental fluorosis or mottled enamel, and elevated fluoride concentrations (>3.0 mg/L) can cause skeletal fluorosis [14–21]. Fluoride concentrations in volcanic rocks (viz., granites, gneisses, and basalts) range from 300 to

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800 ppm, whereas sedimentary rocks (viz., shales, limestone, and sandstone) range from 50 to 800 ppm. Fluorine concentrations in sedimentary rocks range from 200 ppm (limestone) to 1000 ppm (shales). Fluorine is present in carbonate sedimentary rocks as fluorite. Fluorine concentrations in clastic sediments are higher because micas and illites in the clay fractions contain fluorine. Fluoride is commonly found in volcanic rocks. One of the most soluble gases in magmas is hydrogen fluoride, which is partially released during eruptive activity. Fluorine concentrations in igneous and volcanic rocks range from 100 ppm (ultramafic) to >1000 ppm (alkalic). Fluorite, apatite, mica, amphiboles, clays, and williamite are the minerals that have the most impact on fluoride's hydro-geochemistry. Overall water quality influences mineral solubility, complexation, and sorption/ion exchange processes [22–24].

Kadam et al. 2019 investigated boron and fluoride contamination in groundwater and risk assessment in Western India, finding that children in the study location had a moderate noncarcinogenic risk. Suthar 2009 assessed nitrate contamination in groundwater and associated health risk in a portion of Punjab, India; the findings suggest that nitrate is mostly released by fertilizers and solid waste, and poses a serious health risk to children under the age of five. Adimalla et al. 2018 investigated the occurrence of fluoride and nitrate in groundwater, as well as their health risks, using geochemical mechanisms in southern India. They discovered that fluoride primarily comes from host rock, whereas nitrate comes from fertilizers, posing a greater health risk to children than adults. Wang et al. 2016 investigated the polluting factor nitrate in groundwater. In the north China plain, it was discovered that misuse of fertilizers and wastewater discharge are the main causes. However, in this region only a few research activities have been noted especially on groundwater quality and its risk assessment. Sreedhar et al. 2017 reported that weathering of rock-forming minerals strongly reflects the water chemistry in Thummallapalli area, Kadapa district. Sunitha et al. 2019 conducted an investigation on groundwater chemistry of the Lakkireddypalli and Ramapuram region in Kadapa district reported that quality of groundwater mostly depends on mechanisms of silicate weathering and partially carbonate dissolution.

Wu et al. 2016 found that serious pollution present in alluvial plain due to huge population density, intensive manmade activities like industrialization, high water demand, and susceptible environs results in, high health risk to children than adults by oral ingestion due to agricultural and industrial activities in Mid-west China. Kale and Pawar, 2017 found that fluoride enrichment in groundwater is one of the causes of Urolithiasis in a semi-arid region of western India. Pastén-Zapata et al. 2014 analyzed aquifer beneath a sub-humid to humid region in NE Mexico to assess groundwater contamination and found that nitrate isotopes and halide ratios have a variable blend of nitrate sources and changes. The geogenic source for fluoride in groundwater is the occurrence of mineral fluorite in geological formation underlying the area. Apart from this, the fluoride concentration is also elevated due to use of agrochemicals and other alike anthropogenic activities. In addition, it has been detected in areas where high calcite deposition occurs in alkaline situations [25–28].

Nitrate contamination in groundwater has been recognized not only in India, but all throughout the world [29–31]. Higher nitrate concentrations in groundwater were detected in many parts of India, including Andhra Pradesh, Bihar, Delhi, Haryana, Himachal Pradesh, Karnataka, Kerala, Madhya Pradesh, Maharashtra, Orissa, Punjab, Tamil Nadu, Rajasthan, West Bengal, and Uttar Pradesh. Many portions of Andhra Pradesh districts, including Kadapa, Prakasam, Ongole, Guntur, Kurnool, and Ananthapuramu, have higher nitrate concentrations. USEPA established a maximum contamination limit (MCL) of 10 mg/L $\text{NO}_3\text{-N}$ (50 mg/L NO_3), and several research studies reveal that agricultural activities are the principal driver of increased nitrate levels in groundwater. Methemoglobinemia is the most well-known nitrate related health condition, a lethal disease marked by cyanosis and caused by bacteria in the digestive tract converting nitrate (NO_3^-) to nitrite (NO_2^-). It's worth noting that children are particularly vulnerable to this sickness. Another possible nitrate related health consequence is cancer. Although commercial fertilizers are assumed to be the main source of nitrate in groundwater, researchers have discovered that natural organic nitrogen,

livestock, septic tanks, and atmospheric inputs also contribute to contamination [32–36].

The pollution index of groundwater (PIG) is an effective technique for evaluating the suitability of drinking water quality in any area and communicating the overall water quality information. As significant concern has been toxic compounds like fluoride and nitrate are discharged into groundwater, where they can accumulate in crops and have a significant impact on human health [37,38]. The main objective of the study is to assess health risks of fluoride and nitrate in hard rock terrain of Cuddapah, A.P. South India using Pollution Index of Groundwater (PIG).

2. Materials and methods

2.1. Description of the study area

The hard rock area of Y.S.R District, A.P. is located between north latitude $14^{\circ}18'0''$ – $14^{\circ}7'30''$ and east longitude $78^{\circ}20'30''$ – $78^{\circ}49'30''$, with a total area of 510.272 sq.km and is covered by Survey of India toposheet numbers 57 J/16 and J/12 (Fig. 1). Granite, quartzites, and dolomite make to the geology of the research region. The soil is primarily red in colour and is usually of poor quality and barely reaches a few inches below ground level. Weathered shales, fractured shales, and phyllites follow the top soil, which has a thickness of 1–2 m. Various litho unit types from the late Archaean or Early Proterozoic eras cover the terrain, which is followed by rocks from the Dharwarian era, which are both perforated by dolerite dykes. A geological map is shown in Fig. 1. Cuddapah older rocks are superposed by super groups of rocks and the Kurnool group of middle-upper proterozoic age. Among the lithological units encountered are granites, granodiorites, granite-gneiss, quartzites, shales, limestones, dolerite dykes, and quartz reefs. Alluvium, which is made up of gravel, sand, silt, and clay, covers the Pennar River's channels. The top zone in the research area can be up to 10 m thick. The yearly average rainfall during the Southwest monsoon (June to September) is 759 mm. Red soil accounts for 53% of all cultivated land. The majority of ground water in Archaean Crystalline rocks is made up of granite gneisses and migmatites, and it lacks primary porosity. However, the presence of ground water is possible due to the formation of secondary porosity as a result of weathering and fracturing. Ground water occurs in both unconfined and semi-confined circumstances in weathered portions and joints and fractures. Large diameter (6 m) dug wells and dug-cum-bore wells are used to develop ground water in the weathered zone. In most of the area, the weathered zone can be up to 10 m thick. During the wet season, these wells pump for roughly 4 h per day in two periods, yielding 20 to 60 cubic meters per day. During the Rabi season, however, most wells are likely to yield less, ranging from 5 to 20 cum/day, and in drought years, they are likely to dry up. Construction of shallow/deep bore wells is used to develop ground water in the fractured area. The Central Ground Water Board has conducted ground water research at depths ranging from 107.20 to 200 m below ground level. Fractures zones were discovered at depths ranging from 8.00 to 145.80 m bgl, according to the drilling data. However, possible fractures were mostly seen between 20 and 100 m bgl. Fracture zone cumulative yield ranges from 0.1 to 4.9 lps. However, depending on the fractured zone encountered in the bore well, general yield bore wells are reported to be between 1 and 3 lps [32].

2.2. Methods

Thirty groundwater samples were collected in and around hard rock terrain in Cuddapah, A.P. in September 2019, using a random sampling method and taking into account land use, geomorphologic, and topographic aspects of the study area. In preparation for further examination, the samples were collected in polyethylene containers (1-L capacity) and transported to the laboratory, where they were kept in a refrigerator at a temperature below 4 °C. In the field, pH and EC were measured using pH and electrical conductivity (EC) meters. The samples location coordinates were acquired using a Garmin GPS and then exported to the Arc-GIS programme for base map development. The various physicochemical characteristics were determined

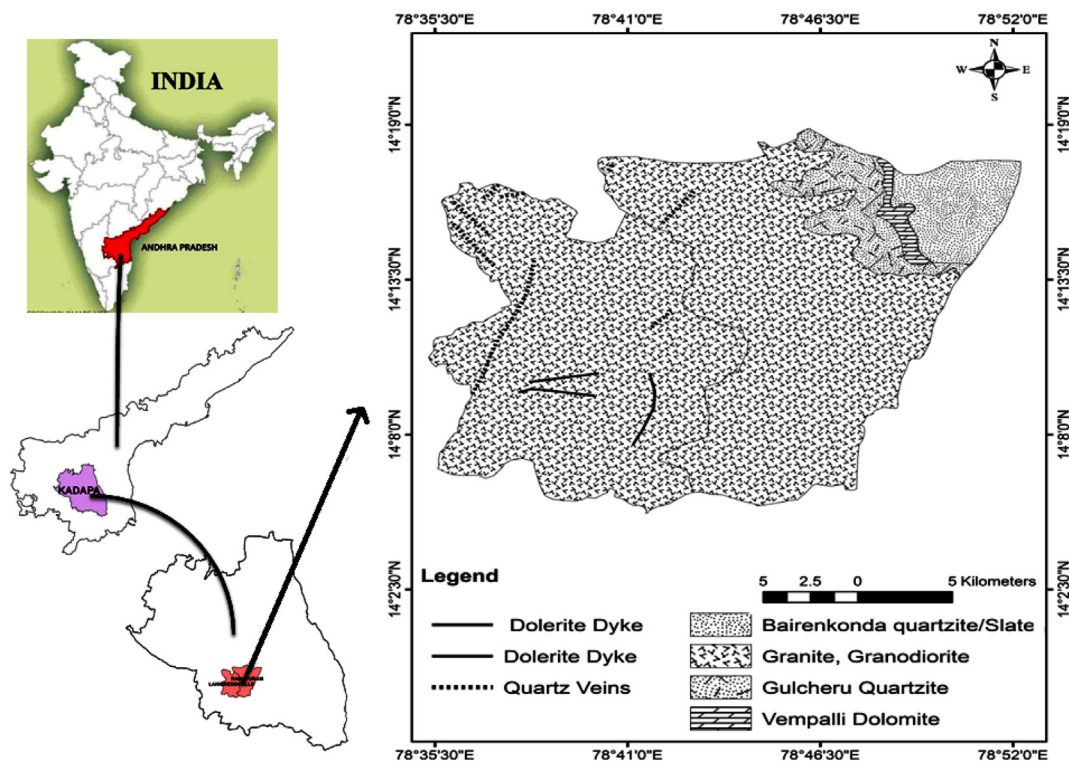


Fig. 1. Location and geological map of the study area.

using standard analytical techniques. Following conventional techniques, the pH, electrical conductivity (EC), total dissolved solids (TDS), major cations, and anions of the water samples were determined [39–41]. A pH meter and a conductivity meter were used to determine pH and EC, respectively. TDS was measured with a TDS meter. An ion selective electrode (Orion 4 star ion meter, Model: pH/ISE) was used to determine F^- . Titrimetry was employed to identify anions and cations, while an ion selective electrode (Orion 4 star ion meter, Model: pH/ISE) was used to determine F^- stock solution (100 mg/L) was used to make standard fluoride solutions (0.1–10 mg/L). To achieve the experimental criteria, 2 ml of total ionic strength adjusting buffer grade III (TISAB III) was added to 20 ml of material. A UV–visible spectrophotometer was used to quantify nitrate (NO_3^-). In groundwater research, a chemical balance error (CBE) of less than 5 is acceptable. The WHO's standard drinking limits were compared to the analytical results of physicochemical parameters [42,43].

2.3. Pollution index of groundwater (PIG) estimation

The pollution index considers the relative importance of each water quality indicator, such as pH, EC, TDS, TH, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , HCO_3^- , SO_4^{2-} , NO_3^- , F^- , and others. A range of procedures were used to generate the groundwater contamination index (PIG).

The Pollution Index of Ground Water (PIG) was first proposed by Subba Rao, 2012. The pollution index takes into consideration the relative influence of each water quality indicator viz., pH, EC, TDS, TH, Ca^{2+} , Mg^{2+} , Na^+ , K^+ , Cl^- , HCO_3^- , SO_4^{2-} , NO_3^- , F^- . The groundwater pollution index (PIG) was calculated using a series of processes [44,45].

The first step is to give each item a weight: As shown in Table 4, individual chemical characteristics have been given a weight (Rw) ranging from 1 to 5 based on the relative relevance of ground water.

The weight parameter is included in the second step: The following equation determines the weight parameter (Wp):

$$Wp = \frac{Rw}{\sum Rw}$$

Rw is the weight of each ingredient, while Wp is the weight parameter. Table 4 shows the determined weight parameter (Wp) values for each ingredient.

In the third stage, the concentration status for each parameter is determined by dividing the individual chemical variable concentration of each water sample by its corresponding drinking water standard quality.

$$Sc = \frac{C}{Ds}$$

where Sc signifies the concentration status, C means the chemical variable concentration of individual water sample, and Ds denotes the standard drinking water level for each chemical constituent.

The overall water quality is considered in the fourth phase: the Wp and Sc are multiplied to obtain the overall water quality (Ow).

$$Ow = Wp \times Sc$$

where Ow is for overall water quality, Wp stands for weight parameter, and Sc stands for concentration status.

The total chemical quality of water is adjusted for across all water quality measures of each water sample in the fifth stage to determine the pollution index of ground water.

$$PIG = \sum Ow$$

when calculating PIG, the contribution of water quality concentration variables of individual water samples was taken into account. Pollution has an impact on groundwater quality if the overall water quality (Ow) reaches 0.1 and accounts for 10% of the PIG's 1.0 value.

2.4. Human health risk assessment

Due to greater knowledge of human health risks, groundwater quality has been steadily decreasing in recent decades as a result of excessive application of fertilizers and pesticides. The assessment of health risks in groundwater is crucial.

In view of extensive application of fertilizers, chemicals, quality of groundwater is continuously deteriorating in recent decades due to increased awareness of health risk on humans [38]. In groundwater evaluation, health risk assessment is critical. Health assessment guarantees that unfavorable human health effects on newborns, children, and adults are identified and evaluated [46,47]. The present work emphasizes the assessment of health risk with reference to NO_3^- and F^- in drinking water in hard rock terrain of Cuddapah, A.P. The chronic daily intake (CDI), hazard quotient (HQ) and total hazard index (THI) were determined using the following formulae:

$$\text{CDI} = \frac{\text{CPW} \times \text{IR} \times \text{ED} \times \text{EF}}{\text{ABW} \times \text{AET}}$$

$$\text{HQ} = \frac{\text{CDI}}{\text{RfD}}$$

$$\text{THI} = \sum \text{HQ}_{\text{fluoride}} + \text{HQ}_{\text{nitrate}}$$

2.4.1. Chronic daily intake (CDI)

To determine the health risk of chronic daily use, the following equation was used:

$$\text{CDI} = \frac{\text{CPW} \times \text{IR} \times \text{ED} \times \text{EF}}{\text{ABW} \times \text{AET}}$$

where CDI stands for chronic daily intake (mg/kg/day), CPW stands for nitrate and fluoride pollution in groundwater (mg/L), and IR stands for ingestion rate (IR) of 2.5 l per day for adults, 0.78 l per day for children, and 0.3 l per day for babies. ED stands for exposure duration years 64, 12 and 1 for adults, children, and infants, respectively [48]. The exposure frequency is set at 365 days per year for adults, children, and infants.

AET means for average exposure duration in days: 23360 for adults, 4380 for children, and 365 for infants ($\text{AET} = \text{ED} \times 365$). ABW represents for average body weight in kilograms: 65 for adults, 20 for children, and 10 for newborns.

2.4.2. Hazard Quotient (HQ)

The following equation was used to quantify the health risk of fluoride and nitrate:

$$\text{HQ} = \frac{\text{CDI}}{\text{RfD}}$$

where HQ denotes hazard quotient, CDI denotes chronic daily intake, and RfD denotes that NO_3^- (1.6 mg/kg/d) and F^- (0.06 mg/kg/d) were obtained from the Integrated Risk Information System (IRIS) data source and the USEPA, respectively (2012).

3. Results and discussions

3.1. Groundwater geochemistry

Many types of geochemical equilibrium use pH to classify them. The pH of groundwater varies from 7.6 to 8.5, indicating that it is alkaline. All of the samples' pH levels are within acceptable ranges. At 25 °C, groundwater's electrical conductivity (EC) ranges from 1100 S/cm to 3870 Siemens/cm, indicating a higher concentration of salts. The standard allowed limit of 1500 S/cm is exceeded in 73.5% of samples. Higher EC could be attributable to high mineral content and increased salinity at the study site, as well as temperature, ion types, and the fact that EC concentration raises with TDS concentration. Higher electrical conductivity variations could possibly be caused by geochemical processes and anthropogenic activities in this area. Total dissolved solids range from 645.4 mg/L to 2400 mg/L (Table 1), with 60% of samples surpassing the permitted range and being unfit for human consumption. Total dissolved solids in water comprise all inorganic salts, such as carbonates, bicarbonates, chlorides, fluoride, sulphates, nitrates, calcium, magnesium, sodium, and potassium [49]. TDS concentrations can be high due to anthropogenic sources such as sewage percolation into groundwater, salt leaching from soil, septic tanks, and agricultural operations. Total hardness ranges from 28 to 540 mg per liter. Only 6.6% of the samples exceeded the permitted limit (Table 1). The concentration of sodium in groundwater varies from 120.8 to 510 mg/L. The permitted limits of 200 mg/L are exceeded in 50% of groundwater. The increase in Na^+ is related to the leaching of the deposit and the breakdown of certain minerals, when such water comes into contact with igneous rocks, sodium gets dissolved from its natural source. Increased salt levels in drinking water, on the other hand, cause hypertension, arteriosclerosis, oedema, hyperosmolarity, renal, and neurological system diseases in humans. High sodium levels are not recommended for agricultural techniques since they degrade the soil. Industrial and municipal waste releases, as well as runoff from diffuse sources, can all contribute to sodium entering natural rivers. The potassium (K^+) concentration in groundwater varies between 74 and 380 mg/L. Carbonate (CO_3^{2-}) levels in groundwater are within legal limits, but bicarbonate (HCO_3^-) levels range from 12 to 38.8 mg/L and 44.8 to 339 mg/L, respectively (Table 1) Most of the groundwater samples are below the 300 mg/L standard. The concentration of chlorides in groundwater varies between 85.4 and 219.7 mg/L.

Higher chlorides are commonly used as a pollution indicator and as a tracer for groundwater contamination. Chloride in groundwater can come from a variety of sources, including weathering, leaching of rocks and soil, and home and municipal effluents. Higher chloride levels are frequently employed as a pollution indicator and a groundwater contamination tracer. Higher chloride concentrations are also generated by household, municipal, and industrial contaminants draining into top soil layers, as well as a dry climate [50,51]. Sulfates in this region range from 23.2 to 110.8 mg/L, showing that the majority of the groundwater is below the 200 mg/L limit. Calcium and magnesium are important

Table 1
Statistical analysis and comparison of groundwater quality parameters with WHO Criteria.

Parameter	Expressed	Minimum	Maximum	W.H.O (2011)		N° of samples exceeding the permissible limit (W.H.O 2011)
				Desirable Limit	Permissible Limit	
pH	Units	7.6	8.5	6.5–8.5	9.2	0
EC	$\mu\text{S}/\text{cm}$	1100	3700	750	1500	22
TDS	mg/L	645.4	2400	500	1500	18
TH	mg/L	28	540	100	500	2
Na^+	mg/L	120.8	510.4	50	200	15
K^+	mg/L	74	380	–	–	0
Ca^{2+}	mg/L	18.2	190	75	200	7
CO_3^{2-}	mg/L	12	38.8	–	–	0
HCO_3^-	mg/L	44.2	339.2	300	–	0
Cl^-	mg/L	48.2	940.2	250	600	5
SO_4^{2-}	mg/L	85.4	219.7	2.5	375	0
NO_3^-	mg/L	23.2	110.8	–	45	26
F^-	mg/L	0.1	3.2	0.5	1.5	12

components of bone and the nervous system, among other things, and they also have an impact on the body's metabolic activity. Calcium levels in groundwater range from 18.2 to 190 mg/L. Calcium levels in drinking water are limited to 75 mg/L. Only 3.3% of groundwater exceeds the calcium regulatory limit.

The research area was discovered to be calcium deficient, with only 3% showing calcium enrichment above the maximum ideal range of 75 mg/L, as recommended by WHO (2011) for drinking purposes. The research area was discovered to be calcium deficient, with only 3% showing calcium enrichment above the maximum ideal range of 75 mg/L, as recommended by WHO (2011) for drinking purposes. Calcium deficiency can lead to osteoporosis, tooth decay, nephrolithiasis (kidney stone), rickets, hypertension, and stroke, among other things. Magnesium concentrations ranged from 22.4 to 95.8 mg/L (Table 1). Magnesium levels in groundwater must not exceed 30 mg/L for drinking purposes. The majority of the samples lie within the magnesium standard limit. Nitrates in groundwater ranges from 23.2 to 110.8 mg/L. According to WHO criteria, 86.6% of groundwater samples have nitrate concentrations are within the permitted level (45 mg/L). Fluorides in groundwater range from 0.1 to 3.2 mg per liter. According to WHO guidelines, 40% of groundwater samples have nitrate concentrations over the allowed level (45 mg/L). The main sources of nitrates are N-compounds from human, animal, and agricultural waste. Urea CO (NH₂)₂ and ammonium nitrate NH₄ NO₃ are the most regularly used fertilizers in this area, accounting for nitrates in the groundwater. Fluoride, at 0.8–1.0 mg/L, is beneficial to human health in the prevention of dental cavities and aids in the maintenance of enamel growth in children under the age of eight. When swallowed in amounts >3.0 mg/L, groundwater with greater fluorides >1.5 mg/L causes dental fluorosis and skeletal fluorosis. Fluoride exists in three forms: fluorspar (calcium fluoride) (CaF₂), apatite (calcium phosphate) (Ca₃F (PO₄)₃), and cryolite (Na₃AlF₆) [52].

3.2. Normalized bivariate plots

The chemistry of groundwater is controlled by a number of lithological processes such as weathering and dissolution. The charts of Ca²⁺/Na⁺ versus HCO₃⁻/Na⁺ and Ca²⁺/Na⁺ versus Mg²⁺/Na⁺ demonstrate this (Fig. 2). The chemistry of groundwater is influenced by silicate weathering and, to a lesser extent, carbonate dissolution. As a result, silicate weathering contributes a significant amount of sodium, potassium, calcium, and magnesium. Similar findings were discovered in lithogenic sites rich in silica [53].

3.3. Geochemical characteristics of groundwater

The concentrations of several essential elements, as well as their interrelationships, were studied in order to better understand and demonstrate the

hydrogeochemical processes at work in the aquifer domain as different groundwater facies evolved. Natural water is the solution of important cations such as Ca²⁺, Mg²⁺, and alkaline metals, as well as important anions such as SO₄²⁻, Cl⁻, CO₃²⁻, and HCO₃⁻. Based on the chemical makeup of groundwater, this graph provides a clear way to identify and relate various types of water. Piper plots are used to represent the classification of water types, hydrogeochemical facies, and geochemical processes that exist inside an aquifer. The hydro geochemistry regime of groundwater is depicted in a trilinear diagram by showing significant cations and anions [54,55]. The Piper diagram is a well-known tool for identifying groundwater types that have a substantial impact on groundwater chemistry. Ca²⁺-HCO₃⁻, Na⁺-Cl⁻, Mixed Ca²⁺-Na⁺-HCO₃⁻, Mixed Ca²⁺-Mg²⁺-Cl⁻, Ca²⁺-Cl⁻, and Na⁺-HCO₃⁻ are the six varieties of groundwater in this region, according to this classification. The majority of the samples, according to the Piper's plot (Fig. 3), are of the Mixed Ca²⁺-Mg²⁺-Cl⁻ type.

3.4. Mechanism controlling the groundwater chemistry

To comprehend the mechanism of groundwater chemistry, the Gibbs (1970) diagram has been employed. It examines chemical reactions utilizing graphical plots of water composition and the regulatory mechanism of natural water chemistry changes. For anions (Cl⁻)/(Cl⁻ + HCO₃⁻) and cations (Na⁺ + K⁺)/(Na⁺ + K⁺ + Ca²⁺) of the groundwater samples, Gibbs (1970) recommended separate plotting of the groundwater samples against TDS. TDS versus Cl⁻/Cl⁻ + HCO₃⁻ and TDS versus (Na⁺ + K⁺)/(Na⁺ + K⁺ + Ca²⁺) were classified into three groups: precipitation zone, rock dominance zone, and evaporation dominance zones. Fig. 4 depicts the plotting of all groundwater samples. The statistics demonstrate that all of the groundwater samples were in the rock dominance (RD) zone, indicating that the principal mechanisms of fluoride release into groundwater are rock weathering and dissolution of fluoride-bearing minerals from the source rock [56].

3.5. Occurrence of fluoride in groundwater

Fluoride (F) is found in trace amounts in natural materials and is also distributed in the air. The earth's crust contains roughly 0.32% fluoride with higher concentrations in lithological compositions including ultramafic rocks (100 mg/kg fluoride), igneous rocks (1000 mg/kg fluoride), and marine shale (1300 mg/kg fluoride). It occurs more frequently in deep groundwater than in shallow water. The most common fluoride concentration-rich minerals in rocks and sediments are fluoroapatite (Ca₅ (PO₄)₃F), topaz (Al₂(SiO₄)F₂), cryolite (Na₃AlF₆), fluorite (CaF₂), and villiaumite (NaF). The presence of quartz granites, monzonites, alkaline volcanic felsic, granodiorites, biotite gneiss, and syenites in geological

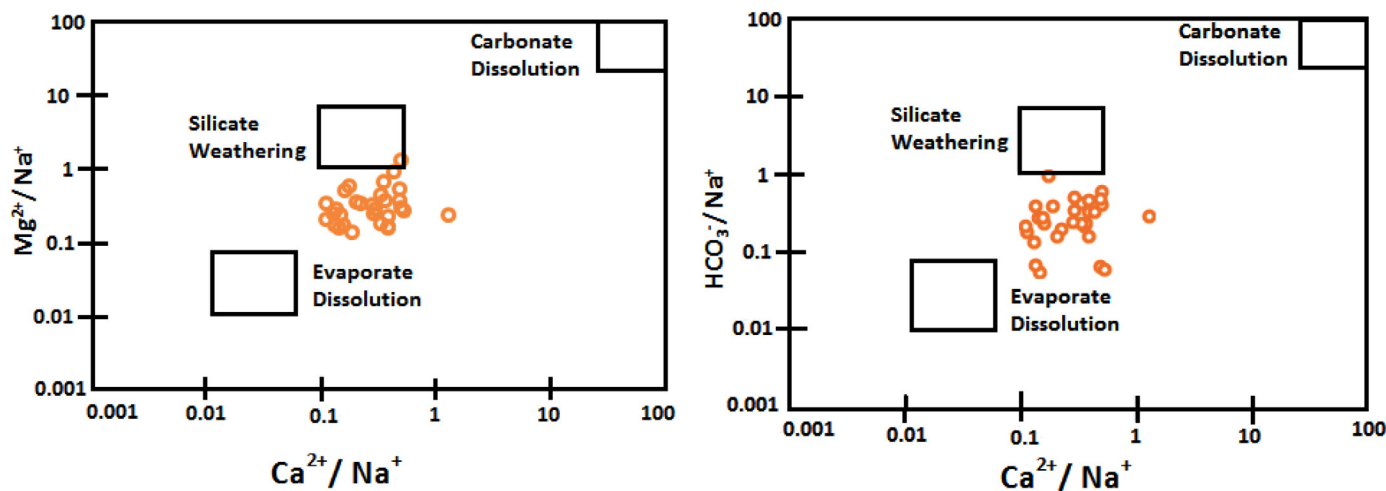


Fig. 2. Bivariate plots diagram.

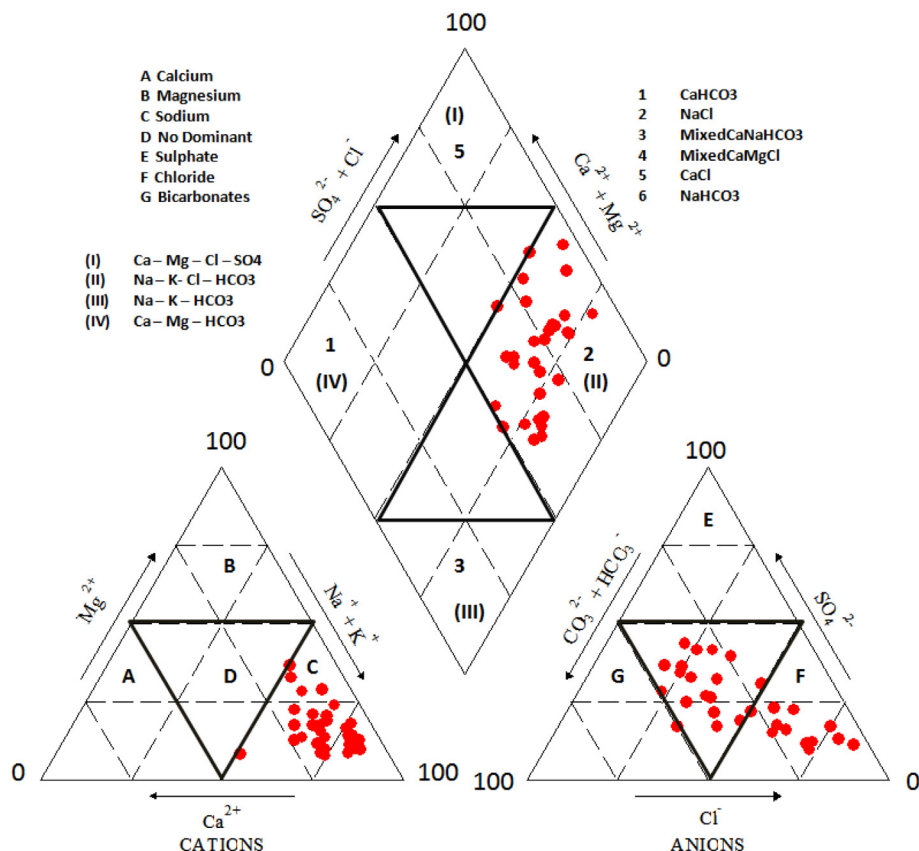


Fig. 3. Piper diagram.

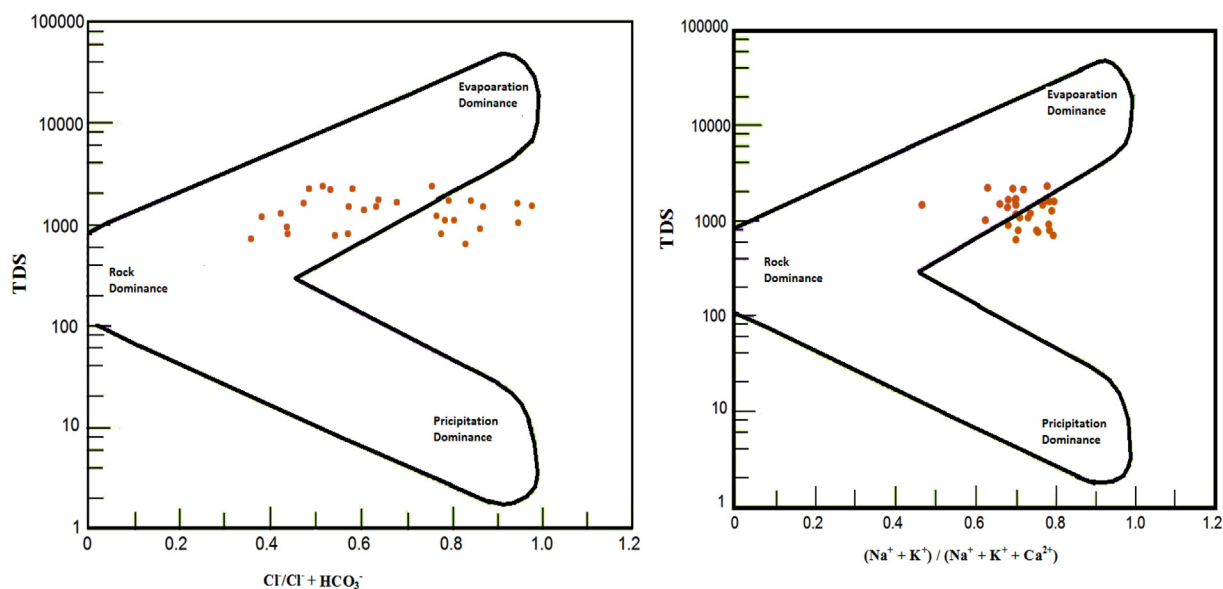


Fig. 4. Mechanism control factor Gibbs (1970) diagram.

formations causes higher amounts of fluoride ions in groundwater [57,58]. Fluorides in this region groundwater range from 0.1 to 3.2 mg/L. Fluoride concentrations in drinking water must be between 0.6 and 1.5 mg/L. Fluoride levels that are less than the minimum suggested limit (0.6 mg/L) induce dental caries, whereas fluoride levels that are more than the maximum advised limit (1.2 mg/L) produce fluorosis. This upper limit prevents tooth decay and promotes healthy bone growth.

About, 60% of groundwater samples are within permissible limit (0.6 to 1.5 mg/L) and 40% of samples exceeded the prescribed level of 1.5 mg/L, resulting in dental fluorosis. Fluoride leaches out as groundwater percolates through rocks containing fluoride-rich substances, and concentrations can rise considerably above the acceptable level. Fluoride content in rocks ranges from 0.1 to 1.0 g/kg on average. Fluorspar, cryolite, and apatite are all important fluoride-containing minerals. F^- on the other hand,

is usually linked with micas and other clay rich minerals in most soils, making it less soluble and hence less hazardous. Fluoride contamination occurs naturally as fluoride-bearing rocks crumble and break down, but if the aquifer's chemistry is altered, the process can be accelerated.

Groundwater is the only available source of drinking water in the research region. The research area was dominated by peninsular gneisses of archaic age rocks, which provided a bountiful supply of fluoride-rich minerals. Fragmented hard rock zones with pegmatite veins are major sources of fluoride in groundwater [59]. High sodium, potassium, bicarbonate, and an alkaline pH promote fluoride ion exchange by hydroxyl ions (OH^-). In semi-arid climatic wet and dry conditions, the Na^+ obtained by groundwater as a result of chemical weathering of rock forming minerals is employed as a metric of mineral weathering. This is what causes fluoride to leach from minerals in soils and rocks. Granite rocks are a natural supply of fluoride, with an average fluoride concentration of 810 mg/kg, and it is thought that granitic terrain groundwater is more conducive to higher fluoride concentrations [60]. In this region's alkaline environment, weathering and leaching of fluoride-bearing minerals from the basement granitic rocks could be a source of fluoride in groundwater. The geological formation is revealed to be the primary source of fluoride concentration in the majority of sampling points. In this approach, geogenic sources (fluoride-bearing minerals such as fluorite, apatite, biotite, hornblende, cryolite, villiaumite, topaz, amphibole, and micas) are the primary determinants of fluoride content in groundwater under alkaline conditions. Fluoride is dissolved in groundwater in India, primarily from geological sources. The major method by which fluoride-rich minerals are decomposed/dissociated from the source rock and fluoride is dissolved in the groundwater by dissolution is rock-water contact. Fluoride ion exchange by hydroxyl ions is increased by high sodium, potassium, bicarbonate, and an alkaline pH. (OH^-). The Na^+ acquired by groundwater as a result of chemical weathering of rock forming minerals is used as a measure of mineral weathering in semi-arid climatic wet and dry circumstances. This is responsible for the leaching of fluoride from minerals in soils and rocks. Geogenic (fluoride-bearing minerals: fluorite, apatite, biotite, hornblende, cryolite, villiaumite, topaz, amphibole, and micas) sources are thus the primary determinants of fluoride content in the study area's groundwater under alkaline conditions. Seasonal, regional, and depth oscillations in fluoride concentration in groundwater are all possible causes of fluoride concentration variations. Furthermore, phosphate fertilizers, which include fluorine as an impurity and are leached down to the saturated zone by irrigation return flows, could contribute a significant quantity of fluorine in addition to geogenic sources [61]. As a result, additional fluoride could be added to the groundwater in the study area.

3.6. Causes and occurrence of nitrate

Domestic/agricultural animal dung, sewage line leakage, landfill leachate, poultry waste disposal, septic tank leaks, and fertilizers used on agricultural regions are all potential causes of nitrate pollution. Excessive nitrate consumption, especially in neonates, might be harmful. Side effects include methemoglobinemia (blue baby syndrome), hypertension, birth anomalies, goiter, and stomach cancer. Infants are especially vulnerable to this illness, which develops when they are fed milk that has high quantities of nitrate. Excessive consumption of nitrate-rich groundwater during pregnancy is harmful to the fetus [62]. Increased nitrate levels in groundwater tests have been linked to the use of chemical fertilizers and waste disposal leachate.

Nitrate contamination is primarily due to extensive use of N-fertilizer and manures, which was the principle cause for higher nitrate concentration in the study area groundwater. Also, this contamination can potentially originate from agricultural fertilizer leaching, septic tank leakage, inappropriate removal of domestic wastes and effluent of organic matter. Consequently, nitrate has the highest penetration depth into soil horizons and finally into the groundwater resources. Excessive nitrate consumption can be dangerous, especially for newborns. Methemoglobinemia (blue baby syndrome), hypertension, birth abnormalities, goiter, and stomach cancer

are all possible side effects. Infants are particularly susceptible to this condition, and this syndrome develops when fed milk includes high levels of nitrate. During pregnancy, excessive intake of nitrate-rich groundwater poses a risk to the baby [63]. The usage of chemical fertilizers and waste disposal leachate has resulted in greater nitrate levels in groundwater samples. The concentration of nitrates in groundwater in this region ranges from 23.2 mg/L to 110.8 mg/L. According to WHO criteria, Nearly 86.6% of groundwater samples have nitrate concentrations lie within the permitted level (45 mg/L). The samples collected from a total of 11 villages have nitrate contents were exceeding the permissible limits.

3.6.1. Nitrate contamination sources

The presence of nitrate in groundwater is due to a variety of sources and origins. The elevated amounts of nitrate in the region's groundwater are attributed to anthropogenic activities such as house sewage leaks and poultry waste dumping, as well as farm animal faeces and nitrate migration from sewage, municipal, and industrial waste. Nitrate can enter groundwater through two different routes: geogenic and anthropogenic activity [64]. In this location, anthropogenic activities are the primary source of nitrate pollution. The origins of increased nitrate in groundwater can be inferred using water samples collected from a variety of locations, including the house, irrigation, and industrial areas. In the research area, inactive/abandoned mines can be located near inhabited areas, in addition to agricultural land regions. Two significant fertilizer components for enhancing paddy crop yields are nitrogen and potassium. Every 2–3 weeks, 200 kg/h of nitrate and potassium fertilizers are applied in this region for paddy production throughout the crop cycle.

Nitrogenous fertilizers are frequently utilized for crop production, which could be a significant source of high NO_3^- concentrations in the groundwater in the study area. As a result, those who drink contaminated water are at danger of suffering a variety of illnesses. Nitrate levels in drinking water have been connected to cancer and have been associated to methemoglobinemia in newborns. Nonpoint nitrate sources, such as chemical fertilizers, and point nitrate sources, such as septic tanks and sewer systems, have been classified by many environmental scientists and researchers. However, during the field investigation, interaction with the residents and personal observation revealed that nitrogenous fertilizers are commonly utilized for irrigation in the region. It has been observed that nitrogenous fertilizers are widely used for crop output, which could be a primary source of high NO_3^- concentrations in the research area's groundwater. As a result, those who drink polluted water are at risk of developing a variety of health problems. High levels of nitrate in drinking water not only induce methemoglobinemia in babies, but they have also been linked to cancer [65]. Many environmental scientists and researchers have classed nitrate sources as nonpoint, such as chemical fertilizers, and point, such as septic tanks and sewer systems. Interaction with the inhabitants and personal observation during the field inquiry, however, revealed that nitrogenous fertilizers are widely used for irrigation in the region. As a result, the study area's high nitrate content is mostly due to anthropogenic causes such as agriculture, home sewage, and septic tank leakage.

3.7. The implications of land use and land cover dynamics on groundwater quality

The quality and quantity of groundwater resources are affected by land use/land cover (LULC) dynamics, particularly expanding urbanization paired with high household wastewater discharge and shrinking open spaces. Groundwater quality variations in 2019, their relationship to land use/land cover, and their influence on local residents in the research region are all investigated in this study. Remote sensing LISS III (30 m spatial resolution) has been applied to quantify land use/land cover changes. The 30-m resolution image was acquired on April 15, 2019. The images were processed and the training sample polygon supervised maximum likelihood classification has been used to classify land use/land cover dynamics. According to classified data, the area of deciduous forest is 437.9 km^2 (0.56%), the construction area is 66.2 km^2 (17.86%), the area of

agricultural land is 36.4 km² (47.24%), the area of scrubland is 145.1 km², vegetation and granite quarry area they are 72.2 and 0.62 km² respectively (Fig. 5).

The effect of land use/cover on water quality varies depending on the kind of terrain. For example, in extensive agriculture, large farms are being cultivated with relatively lower inputs of pesticides, and fertilizers which have not much more impact on groundwater. Small farms are planted in intensive agriculture, which uses a lot of fertilizers and pesticides and has a lot of negative effects on groundwater. In addition, the quality of groundwater on developed land varies widely, depending on the disposal of polluted water utilized for anthropogenic activities and contamination sites. For all water quality metrics, the zonal statistic tool was utilized to estimate the relationship between different land classes and mean concentration changes. According to the LULC categorization, built-up and agricultural land make up the majority of the study area, implying that nitrate contamination may have resulted from the region's intensive irrigation and sewage practices. This region has a total area of 36.4 km² of agricultural land (47.24%).

The LULC analysis showed that the deciduous forest covered most of the study area. The entire forest area is covered with igneous rock composition. These rocks contain a high percentage of fluorine-containing minerals. Due to the water-rock interaction process, these minerals will dissolve in the groundwater. Hence land use practices might also be the source for higher fluoride in groundwater.

3.8. Estimation of pollution index of groundwater (PIG)

PIG is a valuable tool for assessing the quality of water (surface and groundwater), and many researchers have utilized it [66]. According to PIG's classification (Table 2), 23% of the groundwater samples fall into the low contamination zone (PIG: 1.38), 23% of the total groundwater samples fall in the low pollution zone (PIG: 1.38) and 47% of the samples belong to Moderate pollution (PIG 1.83) and 30% of groundwater samples belongs to High Pollution (PIG 2.20).

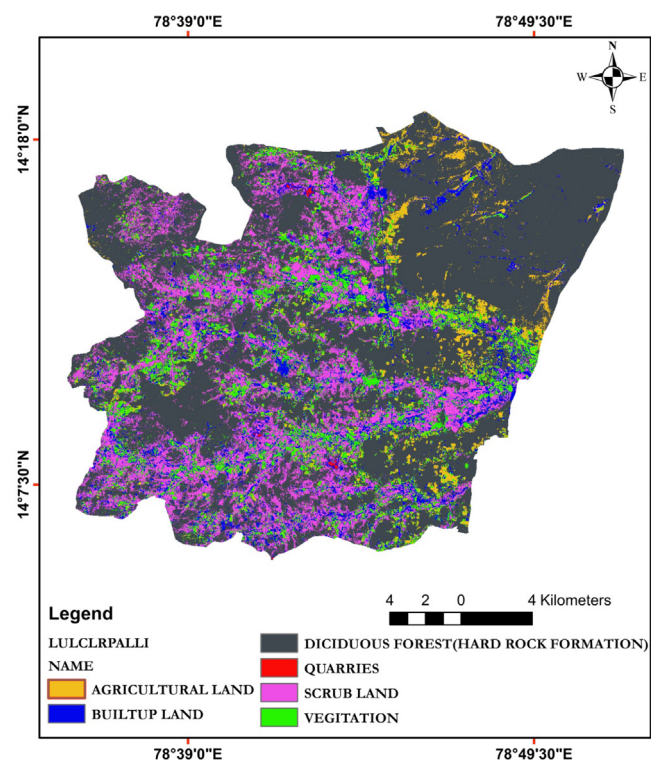


Fig. 5. Land use and Land cover map of the study area.

Table 2

Computed pollution index of groundwater values of the study area.

S.No	PIG	Classification	S.No	PIG	Classification
1	1.69	Moderate pollution	16	1.75	Moderate pollution
2	1.73	Moderate pollution	17	2.37	High pollution
3	1.29	Low pollution	18	1.87	Moderate pollution
4	1.38	Low pollution	19	2.08	High pollution
5	1.59	Moderate pollution	20	1.90	Moderate pollution
6	1.78	Moderate pollution	21	1.30	Low pollution
7	1.82	Moderate pollution	22	1.96	Moderate pollution
8	1.95	Moderate pollution	23	1.99	Moderate pollution
9	1.51	Low pollution	24	2.05	High pollution
10	1.45	Low pollution	25	2.20	High pollution
11	2.33	High pollution	26	1.40	Low pollution
12	2.54	High pollution	27	1.37	Low pollution
13	2.01	High pollution	28	1.79	Moderate pollution
14	2.36	High pollution	29	1.67	Moderate pollution
15	2.05	High pollution	30	1.83	Moderate pollution
				Min	1.29
				Max	2.54
				Mean	1.83

The pH (0.09) and TH (0.04), Na⁺ (0.05), HCO₃⁻ (0.05), SO₄²⁻ (0.07), Cl⁻ (0.03), had Ow values are of less than 0.1. These values are regarded as natural groundwater quality contributors.

However the EC (0.21), TDS (0.14) K⁺ (0.43), Mg²⁺ (0.23) F⁻ (0.16) exhibit the values of Ow higher than 0.1 in the moderate pollution zone. While EC (0.46) TDS (0.29), K⁺ (0.43) Mg²⁺ (0.35) NO₃⁻ (0.22) reveal Ow values greater than 0.1 in the High pollution zone, which are greater than natural groundwater quality contributors. As a result, they clearly show the influence of anthropogenic sources. It is critical to analyze the difference in Ow values between the low moderate and high pollution zones in order to validate the involvement of geogenic and anthropogenic origins as sources of dissolved salts on the aquifer system. Table 3 shows that the values of Ow in the instances of pH, EC, Na, and HCO₃⁻ are nearly identical. However Ow values of K⁺ (0.37), (0.43), (0.43) in the low moderate and high pollution zones respectively and Ow values of NO₃⁻ (0.1), (0.12), (0.22) in the low moderate and high pollution clearly signify the sources of anthropogenic pollution. This disparity in Ow levels may be related to differences in the source of pollution in the groundwater system.

3.9. Evaluation of human health risk assessment (HHRA)

In developing countries like India, where the majority of the population, particularly in rural areas, relies on groundwater for drinking, non-carcinogenic risk for fluoride and nitrate in groundwater is a serious issue. For drinking and domestic use, the majority of people in rural areas rely on groundwater. It is generally known that residents in a variety of South Indian districts suffer from illnesses caused by excessive fluorides and nitrates in their drinking water, particularly fluorosis [67]. According to WHO recommendations, the majority of the groundwater in this region contains fluoride and nitrate levels that exceed the acceptable limit. As a result, it is important to analyze the health risk assessment of nitrate and fluoride by measuring HQ, using universal criteria for fluoride and nitrate.

Fluoride HQ values for adults varied from 0.06 to 2.05, whereas HQ values for children and babies ranged from 0.07 to 2.08 and 0.05 to 1.58, respectively, according to this study. Nitrate HQ values for adults were from 0.68 to 2.87, while those for children and babies were 0.69 to 2.91 and 0.52 to 2.21, respectively. The sum of HQ fluoride and HQ nitrate is known as the Total Hazard Index (THI) and is revealed in Table 4. Adults, children, and newborns had THI values ranging from 0.74 to 4.02, 0.75 to 4.08, and 0.57 to 3.1.

The sum of HQ fluoride and HQ nitrate is known as the Total Hazard Index (THI). According to USEPA guidelines, THI limits for non-carcinogenic harm to human health should not exceed 1. If the THI value is greater than one, the risk to human health is not carcinogenic. In this study, 3% of the groundwater sample was under the maximum permissible

Table 3

Qw and PIG average values.

pH	EC	TDS	TH	Na ⁺	K ⁺	Ca ²⁺	Mg ²⁺	HCO ₃ ⁻	SO ₄ ²⁻	Cl ⁻	NO ₃ ⁻	F ⁻	PIG	Pollution Zone	% of Samples
Ow	Ow	Ow	Ow	Ow	Ow	Ow	Ow	Ow	Ow	Ow	Ow	Ow			
0.09	0.20	0.12	0.04	0.05	0.37	0.03	0.08	0.05	0.07	0.03	0.10	0.16	1.38	Low Pollution	23%
0.09	0.21	0.14	0.11	0.07	0.43	0.07	0.23	0.06	0.08	0.17	0.12	0.04	1.83	Moderate Pollution	47%
0.08	0.46	0.29	0.05	0.03	0.43	0.05	0.35	0.04	0.09	0.03	0.22	0.09	2.20	High Pollution	30%

Table 4

Computation of hazard quotient for adults, children and infants.

S-No	HQF	Adults			Children			Infants		
		HQF	HQN	HQT	HQF	HQN	HQT	HQF	HQN	HQT
1	0.64	1.2	1.84	0.65	1.21	1.86	0.5	0.92	1.42	
2	2.05	1.54	3.59	2.08	1.56	3.64	1.58	1.19	2.77	
3	1.41	1.4	2.81	1.43	1.42	2.85	1.09	1.08	2.17	
4	1.41	0.99	2.4	1.43	1	2.43	1.09	0.76	1.85	
5	1.79	1.06	2.86	1.82	1.08	2.9	1.39	0.82	2.21	
6	1.41	1.88	3.29	1.43	1.91	3.34	1.09	1.45	2.54	
7	0.77	0.97	1.74	0.78	0.98	1.76	0.59	0.75	1.34	
8	1.54	0.68	2.22	1.56	0.69	2.25	1.19	0.53	1.71	
9	0.38	0.95	1.33	0.39	0.96	1.35	0.3	0.73	1.03	
10	0.32	1.26	1.58	0.33	1.28	1.6	0.25	0.97	1.22	
11	0.77	1.45	2.22	0.78	1.47	2.25	0.59	1.12	1.71	
12	0.64	1.91	2.55	0.65	1.94	2.59	0.5	1.48	1.97	
13	0.38	1.67	2.05	0.39	1.69	2.08	0.3	1.29	1.58	
14	0.51	1.78	2.3	0.52	1.81	2.33	0.4	1.38	1.77	
15	0.45	1.88	2.33	0.46	1.91	2.37	0.35	1.46	1.8	
16	0.06	0.68	0.74	0.07	0.69	0.75	0.05	0.52	0.57	
17	0.45	1.5	1.95	0.46	1.52	1.98	0.35	1.16	1.5	
18	1.28	1.64	2.92	1.3	1.66	2.96	0.99	1.27	2.26	
19	1.41	1.7	3.11	1.43	1.72	3.15	1.09	1.31	2.4	
20	0.77	1.9	2.67	0.78	1.93	2.71	0.59	1.47	2.06	
21	0.64	1.11	1.75	0.65	1.13	1.78	0.5	0.86	1.35	
22	0.38	2.84	3.23	0.39	2.88	3.27	0.3	2.19	2.49	
23	1.15	2.87	4.02	1.17	2.91	4.08	0.89	2.21	3.1	
24	1.28	2.36	3.64	1.3	2.39	3.69	0.99	1.82	2.81	
25	0.77	2.15	2.92	0.78	2.18	2.96	0.59	1.66	2.25	
26	0.64	2	2.64	0.65	2.03	2.68	0.5	1.54	2.04	
27	1.28	1.9	3.19	1.3	1.93	3.23	0.99	1.47	2.46	
28	0.38	1.38	1.76	0.39	1.4	1.79	0.3	1.06	1.36	
29	1.28	1.67	2.95	1.3	1.69	2.99	0.99	1.29	2.28	
30	0.32	1.18	1.5	0.33	1.2	1.52	0.25	0.91	1.16	
Min	0.06	0.68	0.74	0.07	0.69	0.75	0.05	0.52	0.57	
Max	2.05	2.87	4.02	2.08	2.91	4.08	1.58	2.21	3.1	
Mean	0.9	1.6	2.5	0.9	1.6	2.5	0.7	1.2	1.9	

limit of THI (1) for adults, children, and newborns, while the remaining 97% of the sample locations above the THI limit of 1 (Table 5). Groundwater from such areas should not be used for drinking or cooking. Groundwater quality can be improved by improving sanitary and drainage systems, as well as implementing safe home and industrial waste disposal systems. Fluoride and nitrate are highly polluted in groundwater, according to the THI risk assessment, and this is a potential cause of health risk in the study area. The current analysis identifies locations of excessive nitrate and fluoride contamination and recommends appropriate solutions to the problem.

Table 5

Assessment of non-carcinogenic risk of adult, children, and infants based on total hazard index.

	*HQ F		HQ N		*THI	
	Safe	Risk	Safe	Risk	Safe	Risk
*NCA	*PS < 1	PS > 1	PS < 1	PS > 1	PS < 1	PS > 1
Adults	18 (60%)	12(40%)	5 (16%)	25 (83%)	1(3%)	29 (96%)
Children	18 (60%)	12 (40%)	4 (13%)	26 (86%)	1(3%)	29 (96%)
Infants	23 (76%)	7 (23%)	10 (33%)	20 (66%)	1(3%)	29 (96%)

* NCA non-carcinogenic risk, HQ hazard quotient, THI total hazard index, NS number of samples, PS percentage of samples.

4. Conclusions

The following are some key findings:

- The research area's groundwater quality is alkaline, with slightly greater total dissolved solids. In this location, the dominant hydro geochemical facies is Mixed Ca⁺-Mg²⁺-Cl⁻.
- 40% of groundwater samples exceeding the maximum allowable fluoride level (1.5 mg/L) and 86% of the samples, have higher the nitrate limit (45 mg/L).
- Rock weathering and dissolution of fluoride-bearing minerals from the source rock are the main contributors to fluoride release in groundwater, and agricultural fertilizers may have contributed to the excessive fluoride in the groundwater. Increased nitrate levels in groundwater samples were also caused by the use of chemical fertilizers and waste disposal leaching.
- A health risk assessment was undertaken since groundwater is not drinkable owing to fluoride and nitrate levels. According to USEPA recommendations, the hazard quotient (HQ) and total hazard index (THI) were determined. Adults had fluoride HQ values of 0.06 to 2.05, children had 0.07 to 2.08, and newborns had 0.05 to 1.58. Adult nitrate HQ values ranged from 0.68 to 2.87, whereas children's nitrate HQ values were from 0.68 and 2.87.
- The suggested THI value for adults, children, and infants is >1. 96% of groundwater exceeds the allowable THI values for adults, children, and infants, according to USEPA rules.
- According to the PIG categorization, 23% of total groundwater samples fall into the low pollution zone (PIG: 1.38) whereas 47% of samples fall into the moderate pollution zone (PIG: 1.83) and 30% of groundwater samples fall into the high pollution zone (PIG: 2.20).

Declaration of Competing Interest

The authors of this article declare that they have no conflict of interests.

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